Genetics of suicides. Family studies and molecular genetics.

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Suicidal behavior, like so much else in psychiatry, tends to cluster in families. Clinical studies show that a family history of suicide is associated with a raised risk of both attempts at suicide and completed suicide. Twin studies show that monozygotic twins have a greater concordance for suicidal behavior than dizygotic twins. Adoption studies also suggest that there may be genetic factors in suicide. Most recently, molecular genetic studies report that polymorphisms of the tryptophan hydroxylase gene are associated with suicidal behavior.

Recent molecular genetic studies and methodological issues in suicide research.

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Suicide behavior (SB) spans a spectrum ranging from suicidal ideation to suicide attempts and completed suicide. Strong evidence suggests a genetic susceptibility to SB, including familial heritability and common occurrence in twins. This review addresses recent molecular genetic studies in SB that include case-control association, genome gene-expression microarray, and genome-wide association (GWA). This work also reviews epigenetics in SB and pharmacogenetic studies of antidepressant-induced suicide. SB fulfills criteria for a complex genetic phenotype in which environmental factors interact with multiple genes to influence susceptibility. So far, case-control association approaches are still the mainstream in SB genetic studies, although whole genome gene-expression microarray and GWA studies have begun to emerge in recent years. Genetic association studies have suggested several genes (e.g., serotonin transporter, tryptophan hydroxylase 2, and brain-derived neurotrophic factor) related to SB, but not all reports support these findings. The case-control approach while useful is limited by present knowledge of disease pathophysiology. Genome-wide studies of gene expression and genetic variation are not constrained by our limited knowledge. However, the explanatory power and path to clinical translation of risk estimates for common variants reported in genome-wide association studies remain unclear because of the presence of rare and structural genetic variation. As whole genome sequencing becomes increasingly widespread, available genomic information will no longer be the limiting factor in applying genetics to clinical medicine. These approaches provide exciting new avenues to identify new candidate genes for SB genetic studies. The other limitation of genetic association is the lack of a consistent definition of the SB phenotype among studies, an inconsistency that hampers the comparability of the studies and data pooling. In summary, SB involves multiple genes interacting with non-genetic factors. A better understanding of the SB genes by combining whole genome approaches with case-control association studies, may potentially lead to developing effective screening, prevention, and management of SB.

Genetic contributions to suicidal thoughts and behaviors.

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Suicidal ideation, suicide attempt (SA) and suicide are significantly heritable phenotypes. However, the extent to which these phenotypes share genetic architecture is unclear. This question is of great relevance to determining key risk factors for suicide, and to alleviate the societal burden of suicidal thoughts and behaviors (STBs). To help address the question of heterogeneity, consortia efforts have recently shifted from a focus on suicide within the context of major psychopathology (e.g. major depressive disorder, schizophrenia) to suicide as an independent entity. Recent molecular studies of suicide risk by members of the Psychiatric Genomics Consortium and the International Suicide Genetics Consortium have identified genome-wide significant loci associated with SA and with suicide death, and have examined these phenotypes within and outside of the context of major psychopathology. This review summarizes important insights from epidemiological and biometrical research on suicide, and discusses key empirical findings from molecular genetic examinations of STBs. Polygenic risk scores for these phenotypes have been observed to be associated with case-control status and other risk phenotypes. In addition, estimated shared genetic covariance with other phenotypes suggests specific medical and psychiatric risks beyond major depressive disorder. Broadly, molecular studies suggest a complexity of suicide etiology that cannot simply be accounted for by depression. Discussion of the state of suicide genetics, a growing field, also includes important ethical and clinical implications of studying the genetic risk of suicide.

Genome-Wide DNA Methylation Patterns in Suicide Victims: Identification of New Candidate Genes.

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Suicide is a major global public health problem with significant impact on society. According to the World Health Organization, every year about 800.000 people commit suicide, while at the global level suicide accounts for 50% of all violent deaths among men and for 71% among women. Suicide is a complex phenomenon which cannot be attributed to a single causal factor, but to a combination of simultaneous effects of multiple factors which are expressed in the form of psychological, biological and sociological indicators. Analysis of epigenetic mechanisms (methylation of the DNA, modifications of histone proteins and (networks of) miRNA), which link the interaction between genes and the environment, could importantly contribute to better understanding of suicidal behaviour. Recent studies on suicidal behaviour and DNA methylation show differences in DNA methylation pattern, with numerous sites among suicide victims. Using next generation sequencing, genome-wide studies helped identify novel candidate genes while studies of already known candidate genes (such as glucocorticoid receptor and BDNF) gave us better insight into the interplay of genetics and epigenetics. Epigenetic studies importantly contribute to elucidation of new biomarkers for suicidal behaviour. However, present studies are very different in design and often performed on very small samples, and these limitations could be overcome with more careful study preparation.

Recent Breakthroughs in Genetic and Brain Structural Correlates of Suicidal Behaviors: A Short Review.

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Suicide accounts for more than 700,000 deaths annually and is the fourth leading cause of death among individuals aged 15 to 29 years. Despite years of research to understand the etiology and pathophysiology of suicidal behavior, many questions remain unresolved-for example, whether suicidal behavior has a unique genetic or neurobiological basis and how these differ from related psychiatric conditions, such as depression, bipolar disorder, schizophrenia, etc. Identifying these biological correlates is paramount to advancing our understanding of the mechanisms that underlie suicidal behavior. In this literature review, we examine the complex nature of suicidal thoughts and behaviors, integrating insights from large-scale genetic and neuroimaging studies published between 2018 and 2023. Recent genome-wide association studies have uncovered specific genomic loci associated with specific suicidal behaviors. However, there is a need for larger and more diverse samples in these studies to overcome challenges in replicability and generalizability. Neuroimaging studies have also revealed structural brain differences associated with suicidal behavior, thanks to international consortium-level efforts that have enabled data sharing, collaboration, and coordinated analyses that improve the robustness and reliability of findings. Despite promising progress in identifying the genetic and neurobiological underpinnings of suicidal behavior, the translation of these advances and findings into effective suicide prevention strategies and clinical tools remains a crucial challenge. Consequently, future studies must focus on integrating biological elements into an improved mechanistic understanding of the etiology of suicidal behavior, which in turn can translate into new strategies for early detection, intervention, and treatment.

Genome-wide methylome-based molecular pathologies associated with depression and suicide.

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Major depressive disorder (MDD) is a debilitating disorder. Suicide attempts are 5-times higher in MDD patients than in the general population. Interestingly, not all MDD patients develop suicidal thoughts or complete suicide. Thus, it is important to study the risk factors that can distinguish suicidality among MDD patients. The present study examined if DNA methylation changes can distinguish suicidal behavior among depressed subjects. Genome-wide DNA methylation was examined in the dorsolateral prefrontal cortex of depressed suicide (MDD+S; n = 15), depressed non-suicide (MDD-S; n = 17), and nonpsychiatric control (C; n = 16) subjects using 850 K Infinium Methylation EPIC BeadChip. The significantly differentially methylated genes were used to determine the functional enrichment of genes for ontological clustering and pathway analysis. Based on the number of CpG content and their relative distribution from specific landmark regions of genes, 32,958 methylation sites were identified across 12,574 genes in C vs. MDD+/-S subjects, 30,852 methylation sites across 12,019 genes in C vs. MDD-S, 41,648 methylation sites across 13,941 genes in C vs. MDD+S, and 49,848 methylation sites across 15,015 genes in MDD-S vs. MDD+S groups. A comparison of methylation sites showed 33,129 unique methylation sites and 5451 genes in the MDD-S group compared to the MDD+S group. Functional analysis suggested oxytocin, GABA, VGFA, TNFA, and mTOR pathways associated with suicide in the MDD group. Altogether, our data show a distinct pattern of DNA methylation, the genomic distribution of differentially methylated sites, gene enrichment, and pathways in MDD suicide compared to non-suicide MDD subjects.

Electroacupuncture Reduces Fibromyalgia Pain via Neuronal/Microglial Inactivation and Toll-like Receptor 4 in the Mouse Brain: Precise Interpretation of Chemogenetics.

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Fibromyalgia (FM) is a complex, chronic, widespread pain syndrome that can cause significant health and economic burden. Emerging evidence has shown that neuroinflammation is an underlying pathological mechanism in FM. Toll-like receptors (TLRs) are key mediators of the immune system. TLR4 is expressed primarily in microglia and regulates downstream signaling pathways, such as MyD88/NF-κB and TRIF/IRF3. It remains unknown whether electroacupuncture (EA) has therapeutic benefit in attenuating FM pain and what role the TLR4 pathway may play in this effect. We compared EA with sham EA to eliminate the placebo effect due to acupuncture. We demonstrated that intermittent cold stress significantly induced an increase in mechanical and thermal FM pain in mice (mechanical: 2.48 ± 0.53 g; thermal: 5.64 ± 0.32 s). EA but not sham EA has an analgesic effect on FM mice. TLR4 and inflammatory mediator-related molecules were increased in the thalamus, medial prefrontal cortex, somatosensory cortex (SSC), and amygdala of FM mice, indicating neuroinflammation and microglial activation. These molecules were reduced by EA but not sham EA. Furthermore, a new chemogenetics method was used to precisely inhibit SSC activity that displayed an anti-nociceptive effect through the TLR4 pathway. Our results imply that the analgesic effect of EA is associated with TLR4 downregulation. We provide novel evidence that EA modulates the TLR4 signaling pathway, revealing potential therapeutic targets for FM pain.

Neuroplasticity Intervention, Amygdala and Insula Retraining (AIR), Significantly Improves Overall Health and Functioning Across Various Chronic Conditions.

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Chronic conditions, sometimes referred to as functional somatic disorders, such as myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), fibromyalgia (FM), and more recently, long COVID (LC), affect millions of people worldwide. Yet, after decades of research and testing, the etiology and treatment for many of these diseases is still unclear. Recently, a consortium of clinicians and researchers have proposed that while many different chronic conditions exist, the root cause of each may be a similar brain-body connection, as the brain responds to perceived biological threats and transmits danger signals to the body that manifest as somatic symptoms. This hypothesis suggests that treating chronic conditions requires an approach that addresses the neural networks involved. One such method, known as Amygdala and Insula Retraining (AIR), otherwise known as The Gupta Program, has shown promise in recent years for treating such conditions, including ME/CFS, FM, and LC. The present study aimed to demonstrate that AIR could be an effective approach for numerous other chronic illnesses (e.g., Lyme disease, mold illness, mast cell activation syndrome [MCAS]) and others. This novel and exploratory research examined self-reported health and functioning levels before and after using AIR. A series of paired-sample t tests with Bonferroni correction demonstrated that after 3+ months of using AIR (the minimum recommended time for the intervention), participants experienced a significant increase in overall health and functioning for 14 of 16 conditions tested ($P < .001$ for all but one, which was $P = .001$) and approached significance for the remaining two conditions ($P = .039$ and $P = .005$). Of the 14 signficant findings, 11 had a large effect size and three had a medium effect size. Naturally, this study has limitations. It was a cross-sectional design with a small convenience sample and self-reported data. Future research with larger samples and randomized controlled trials is needed to provide further evidence of AIR's effectiveness. Nonetheless, these preliminary findings suggest that AIR is a viable method for improving the health of people suffering from chronic conditions, and clinicians and researchers might consider incorporating AIR into their protocols for these patients.

Healthcare experiences of fibromyalgia patients and their associations with satisfaction and pain relief. A patient survey.

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The etiology of fibromyalgia (FM) is disputed, and there is no established cure. Quantitative data on how this may affect patients' healthcare experiences are scarce. The present study aims to investigate FM patients' pain-related healthcare experiences and explore factors associated with high satisfaction and pain relief. An anonymous, online, and patient-administered survey was developed and distributed to members of the Norwegian Fibromyalgia Association. It addressed their pain-related healthcare experiences from both primary and specialist care. Odds ratios for healthcare satisfaction and pain relief were estimated by binary logistic regression. Directed acyclic graphs guided the multivariable analyses. The patients (n = 1,626, mean age: 51 years) were primarily women (95%) with a 21.8-year mean pain duration and 12.7 years in pain before diagnosis. One-third did not understand why they had pain, and 56.6% did not know how to get better. More than half had not received satisfactory information on their pain cause from a physician, and guidance on how to improve was reported below medium. Patients regretted a lack of medical specialized competence on muscle pain and reported many unmet needs, including regular follow-up and pain assessment. Physician-mediated pain relief was low, and guideline adherence was deficient. Only 14.8% were satisfied with non-physician health providers evaluating and treating their pain, and 21.5% were satisfied (46.9% dissatisfied) with their global pain-related healthcare. Patients' knowledge of their condition, physicians' pain competence and provision of information and guidance, agreement in explanations and advice, and the absence of unmet needs significantly increased the odds of both healthcare satisfaction and pain relief. Our survey describes deficiencies in FM patients' pain-related healthcare and suggests areas for improvement to increase healthcare satisfaction and pain relief. (REC# 2019/845, 09.05.19).

Study protocol for "Psilocybin in patients with fibromyalgia: brain biomarkers of action".

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Chronic pain is a leading cause of disability worldwide. Fibromyalgia is a particularly debilitating form of widespread chronic pain. Fibromyalgia remains poorly understood, and treatment options are limited or moderately effective at best. Here, we present a protocol for a mechanistic study investigating the effects of psychedelic-assisted-therapy in a fibromyalgia population. The principal focus of this trial is the central mechanism(s) of psilocybin-therapy i.e., in the brain and on associated mental schemata, primarily captured by electroencephalography (EEG) recordings of the acute psychedelic state, plus pre and post Magnetic Resonance Imaging (MRI). Twenty participants with fibromyalgia will complete 8 study visits over 8 weeks. This will include two dosing sessions where participants will receive psilocybin at least once, with doses varying up to 25mg. Our primary outcomes are 1) Lempel-Ziv complexity (LZc) recorded acutely using EEG, and the 2) the (Brief Experiential Avoidance Questionnaire (BEAQ) measured at baseline and primary endpoint. Secondary outcomes will aim to capture broad aspects of the pain experience and related features through neuroimaging, self-report measures, behavioural paradigms, and qualitative interviews. Pain Symptomatology will be measured using the Brief Pain Inventory Interference Subscale (BPI-IS), physical and mental health-related function will be measured using the 36-Item Short Form Health Survey (SF-36). Further neurobiological investigations will include functional MRI (fMRI) and diffusion tensor imaging (changes from baseline to primary endpoint), and acute changes in pre- vs post-acute spontaneous brain activity - plus event-related potential functional plasticity markers, captured via EEG. The results of this study will provide valuable insight into the brain

mechanisms involved in the action of psilocybin-therapy for fibromyalgia with potential implications for the therapeutic action of psychedelic-therapy more broadly. It will also deliver essential data to inform the design of a potential subsequent RCT.